Exercise Set 6: Smooth Manifolds

We now add the most important structure of all: smoothness. A topological manifold is a space that locally looks like $\mathbb{R}^n$. A smooth manifold adds a consistent notion of differentiation by requiring that the coordinate changes the transition maps are smooth. This is what allows us to talk about velocity, tangent vectors, and rates of change on curved spaces.

Recall of Definitions and Examples

Charts and atlases

R1. Write the definition of a **chart** (U, ϕ) on a topological space M. What are the three properties it must satisfy? Draw a picture showing a chart as a "coordinate patch" on a manifold.

R2. Write the definition of an **atlas** $\mathcal{A} = \{(U_\alpha, \phi_\alpha)\}$ on M. What single covering condition must the collection of charts satisfy?

R3. Write the definition of a **smooth manifold**. What two conditions does a smooth manifold require beyond being a topological manifold?

(a) The Hausdorff condition why is it needed?

(b) Second countability what does this rule out?

(c) Smooth compatibility of charts state this precisely using transition maps.

R4. Write the definition of the **transition map** $\phi_\beta \circ \phi_\alpha^{-1}$ between two overlapping charts (U_α, ϕ_α) and (U_β, ϕ_β).

(a) What is the domain of this map?

(b) What is the codomain?

(c) Draw a commutative diagram showing how the two charts and the transition map relate.

(d) What does it mean for two charts to be **smoothly compatible**?

R5. Write the definition of a **maximal atlas** (also called a smooth structure). Why do we pass to the maximal atlas? What is the relationship between an atlas and the maximal atlas it generates?

Examples of smooth manifolds

R6. For each of the following, state the dimension and give an explicit atlas with the minimum number of charts needed:

(a) $\mathbb{R}^n$

(b) The open interval $(0, 1)$

(c) S^1 (two charts via angle, or two via stereographic projection)

(d) S^2 (two charts via stereographic projection)

(e) $T^2 = S^1 \times S^1$

R7. For the circle S^1, write out the **stereographic atlas** explicitly:

(a) Define $U_N = S^1 \setminus \{(0, 1)\}$ and the chart $\phi_N : U_N \to \mathbb{R}$ by stereographic projection from the north pole $(0, 1)$. Write the explicit formula for $\phi_N(x, y)$.

(b) Define $U_S = S^1 \setminus \{(0, -1)\}$ and $\phi_S : U_S \to \mathbb{R}$ by stereographic projection from the south pole. Write the formula.

(c) Compute the transition map $\phi_S \circ \phi_N^{-1} : \phi_N(U_N \cap U_S) \to \phi_S(U_N \cap U_S)$. Show it is smooth.

(d) What is $\phi_N(U_N \cap U_S)$? What is $\phi_S(U_N \cap U_S)$?

R8. Write the definition of a **smooth map** $f : M \to N$ between smooth manifolds. How does it use charts? Write the definition of a **diffeomorphism**.

Computation

Equivalent atlases on $\mathbb{R}$

The key insight is that the *same* topological space can carry different smooth structures, but many of them turn out to be equivalent (diffeomorphic).

C1. (Two atlases on $\mathbb{R}$: the standard and the cube root.)

Consider the topological manifold $M = \mathbb{R}$ with two atlases:

$$\mathcal{A}_1 = \{(\mathbb{R}, \phi_1)\}, \qquad \phi_1(x) = x \quad \text{(the identity chart)},$$

$$\mathcal{A}_2 = \{(\mathbb{R}, \phi_2)\}, \qquad \phi_2(x) = x^3 \quad \text{(the cube root chart)}.$$

(a) Show that both ϕ_1 and ϕ_2 are homeomorphisms $\mathbb{R} \to \mathbb{R}$.

(b) Compute the transition map from $\mathcal{A}_1$ to $\mathcal{A}_2$:

$$\phi_2 \circ \phi_1^{-1} : \mathbb{R} \to \mathbb{R}.$$

Is it smooth? Is it a diffeomorphism?

(c) Compute the transition map from $\mathcal{A}_2$ to $\mathcal{A}_1$:

$$\phi_1 \circ \phi_2^{-1} : \mathbb{R} \to \mathbb{R}.$$

Is it smooth at 0? Compute its derivative at 0.

(d) Conclude: the charts ϕ_1 and ϕ_2 are *not* smoothly compatible. Therefore $\mathcal{A}_1$ and $\mathcal{A}_2$ define *different* smooth structures on $\mathbb{R}$.

(e) However, show that the two smooth manifolds $(\mathbb{R}, \mathcal{A}_1)$ and $(\mathbb{R}, \mathcal{A}_2)$ are **diffeomorphic**: find an explicit diffeomorphism $f : (\mathbb{R}, \mathcal{A}_1) \to (\mathbb{R}, \mathcal{A}_2)$.
Hint: what map f makes the coordinate representation $\phi_2 \circ f \circ \phi_1^{-1}$ smooth?

(f) **Key lesson.** Different smooth structures can be defined on the same topological space, but they may still be diffeomorphic. In dimension 1, all smooth structures on $\mathbb{R}$ are diffeomorphic. (In dimension 4, this fails spectacularly there exist "exotic $\mathbb{R}^4$".)

C2. (The angle atlas on S^1.)

Consider the circle S^1 with the following two-chart atlas:

$$U_1 = S^1 \setminus \{(-1, 0)\}, \quad \psi_1(\cos\theta, \sin\theta) = \theta \in (-\pi, \pi),$$

$$U_2 = S^1 \setminus \{(1, 0)\}, \quad \psi_2(\cos\theta, \sin\theta) = \theta \in (0, 2\pi).$$

Call this atlas $\mathcal{A}_{\text{angle}}$.

(a) Describe the overlap $U_1 \cap U_2$. What is $\psi_1(U_1 \cap U_2) \subset (-\pi, \pi)$? It has two connected components describe them.

(b) Compute the transition map $\psi_2 \circ \psi_1^{-1} : \psi_1(U_1 \cap U_2) \to \psi_2(U_1 \cap U_2)$. Write the explicit formula on each connected component. Is it smooth?

(c) Is the transition map a diffeomorphism? Compute its derivative.

C3. (The stereographic atlas on S^1 same smooth structure.)

Recall the stereographic atlas $\mathcal{A}_{\text{stereo}}$ from R7 with charts ϕ_N, ϕ_S.

(a) Show that the chart ψ_1 from $\mathcal{A}_{\text{angle}}$ is smoothly compatible with ϕ_N from $\mathcal{A}_{\text{stereo}}$: compute the transition map $\phi_N \circ \psi_1^{-1} : \psi_1(U_1 \cap U_N) \to \phi_N(U_1 \cap U_N)$ and show it is smooth with smooth inverse.

(b) Since each chart of $\mathcal{A}_{\text{angle}}$ is compatible with each chart of $\mathcal{A}_{\text{stereo}}$, the two atlases are compatible and belong to the **same maximal atlas** i.e. the same smooth structure on S^1.

(c) **Key lesson.** Even though $\mathcal{A}_{\text{angle}}$ and $\mathcal{A}_{\text{stereo}}$ look different, they are just two different descriptions of the same smooth circle. The smooth structure of S^1 is unique (up to diffeomorphism). Write a one-sentence summary of what this means physically.

C4. (Smooth maps in coordinates.)

Let $f : S^1 \to S^1$ be defined by $f(\cos\theta, \sin\theta) = (\cos 2\theta, \sin 2\theta)$ (the double cover).

(a) Using the angle charts ψ_1 on both domain and codomain, write the coordinate representation $\psi_1 \circ f \circ \psi_1^{-1}$. Is it smooth?

(b) At the point $p = (1, 0)$, compute the derivative of the coordinate representation. What does this number measure geometrically?

(c) Is f a diffeomorphism? Justify using the definition.

C5. (Product and quotient manifolds.)

(a) The torus $T^2 = S^1 \times S^1$ is a smooth manifold with the product atlas: if (U_α, ϕ_α) is an atlas for S^1, write down the atlas for T^2 in terms of product charts $U_\alpha \times U_\beta$ and maps $\phi_\alpha \times \phi_\beta$.

(b) Write the transition maps for T^2 in terms of the transition maps of S^1. Are they smooth?

(c) The cylinder $S^1 \times \mathbb{R}$ is also a smooth manifold. Write an explicit atlas for it and verify the transition maps are smooth.

(d) **(The quotient $\mathbb{R}/\mathbb{Z}$).** The circle can also be defined as $\mathbb{R}/\mathbb{Z}$ (identify $x \sim x + n$ for all $n \in \mathbb{Z}$). The quotient map $\pi : \mathbb{R} \to \mathbb{R}/\mathbb{Z}$ allows us to push charts from $\mathbb{R}$ down to the quotient. Describe how to construct an atlas on $\mathbb{R}/\mathbb{Z}$ from this quotient map, and verify the transition maps are smooth. Show that $\mathbb{R}/\mathbb{Z} \cong S^1$ as smooth manifolds.

Physics: Velocity on Configuration Space

In Set 5 we described the positions *a physical system can occupy the configuration space is a smooth manifold. Now that we have smooth structure, we can ask the next question:* how fast is the system moving, and in what direction? *A smooth curve on the configuration manifold has a well-defined velocity at each moment. This is the geometric meaning of a tangent vector and it is why physics needs smooth manifolds, not just topological ones.*

P1. (Velocity of a particle on S^2.)

A particle constrained to move on the unit sphere S^2 traces a smooth curve $\gamma : \mathbb{R} \to S^2$. In the spherical coordinate chart (θ, ϕ), the curve is written as $(\theta(t), \phi(t))$.

(a) The **speed** of the particle in coordinates is

$$v(t) = \sqrt{\dot\theta(t)^2 + \sin^2\theta(t)\,\dot\phi(t)^2}.$$

Why does the speed depend on θ even though the chart uses (θ, ϕ) as independent coordinates? *(Hint: think about what happens near the poles.)*

(b) Consider the curve $\gamma(t) = (\pi/2, t)$ a particle moving along the equator at unit angular speed. Compute its velocity vector in the chart (θ, ϕ). What is its speed?

(c) Now consider $\gamma(t) = (t, 0)$ a particle moving along a meridian. Compute its velocity vector. What is its speed?

(d) At the north pole $p = (0, 0, 1)$, the spherical chart breaks down ($\theta = 0$). Switch to the stereographic chart ψ_N from R7. Write the same curve $\gamma(t) = (t, 0)$ in the stereographic chart near $t = 0$ and compute the velocity there.

(e) **Key point.** The velocity vector exists independently of the chart but its *components* change when you change charts. The transition map tells you exactly how: if $(\phi_\beta \circ \phi_\alpha^{-1})$ is the transition map, the velocity components transform by its Jacobian. This is why we need smooth (not just continuous) transition maps: the Jacobian must exist.

P2. (Velocity on the double pendulum T^2.)

Return to the double pendulum from Set 5. Its configuration space is T^2 with coordinates $(\theta_1, \theta_2) \in [0, 2\pi) \times [0, 2\pi)$. A motion of the system is a smooth curve $\gamma(t) = (\theta_1(t), \theta_2(t))$ on T^2.

(a) The **velocity** of the system at time t is the pair $(\dot\theta_1(t), \dot\theta_2(t))$. This is a tangent vector to T^2 at the point $(\theta_1(t), \theta_2(t))$. What is the dimension of the space of all possible velocities at a given configuration? What does each direction correspond to physically?

(b) Consider the specific motion:

$$\theta_1(t) = \omega_1 t, \qquad \theta_2(t) = \omega_2 t,$$

for constants $\omega_1, \omega_2 > 0$ (constant angular velocities on each rod). Write the velocity vector in the (θ_1, θ_2) chart.

(c) For $\omega_1 = \omega_2 = 1$: the orbit is the diagonal line on the torus. Is this a closed curve on T^2? What is its period?

(d) For $\omega_1 = 1$, $\omega_2 = \sqrt{2}$: you showed in Set 5 that the orbit is dense on T^2. What does the velocity vector look like at each point of the orbit? Is the speed constant?

(e) The **kinetic energy** of the double pendulum is

$$K = \frac{1}{2}(m_1\ell_1^2 + m_2\ell_1^2)\dot\theta_1^2 + \frac{1}{2}m_2\ell_2^2\dot\theta_2^2 + m_2\ell_1\ell_2\dot\theta_1\dot\theta_2\cos(\theta_1 - \theta_2).$$

This is a quadratic function of the velocity vector $(\dot\theta_1, \dot\theta_2)$ specifically, $K = \frac{1}{2}\, g_{ij}\dot\theta^i\dot\theta^j$ where g_{ij} is a 2×2 matrix depending on (θ_1, θ_2).
Write the matrix g_{ij} explicitly. Notice that g_{ij} is not constant it depends on the configuration. This is the **Riemannian metric** on the configuration space. We will define this properly in the next set.

P3. (What smooth structure adds comparing with Set 5.)

In Set 5, the configuration space T^2 was only a topological space. Now it is a smooth manifold. Compare what we can and cannot say in each case.

Question	Topological T^2	Smooth T^2
Can we describe what positions are possible?	Yes	Yes
Can we say whether a motion is continuous?	Yes	Yes
Can we define the velocity of a motion?	No	Yes
Can we compute the kinetic energy?	No	Yes
Can we write the equations of motion?	No	Yes
Can we define the Lagrangian $L = K - V$?	No	Yes

(a) A **topological** path $\gamma : [0, 1] \to T^2$ might not have a velocity anywhere it could be nowhere differentiable. Give an example of such a path explicitly.

(b) A **smooth** path always has a velocity $\dot{\gamma}(t)$ at every t. The smoothness of the manifold (smooth transition maps) ensures this is well-defined independently of the chart.

(c) **The key point.** Physics requires smooth manifolds because the laws of motion are differential equations. To write $F = ma$ or the Euler-Lagrange equations on a manifold, you need to differentiate and differentiation requires smooth charts. A purely topological configuration space is not enough.

P4. (Speed of convergence returning to the running problem.)

In Sets 2 and 3, we studied the orbit of $f : X \to X$ and asked whether it converges. In Set 4, we used compactness to guarantee fixed points exist. Now, with smooth structure, we can ask *how fast* the orbit converges.

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth map with a fixed point x^* (so $f(x^*) = x^*$).

(a) Show that the orbit $x_{n+1} = f(x_n)$ satisfies

$$x_{n+1} - x^* \approx f'(x^*)\,(x_n - x^*)$$

for x_n close to x^*. (Use the Taylor expansion of f at x^*.)

(b) Conclude: if $|f'(x^*)| < 1$, the orbit converges to x^* exponentially fast. If $|f'(x^*)| > 1$, the orbit diverges. If $|f'(x^*)| = 1$, the linear analysis is inconclusive.

(c) For $f(x) = \cos x$: find the fixed point x^* numerically (it satisfies $\cos x^* = x^*$, the "Dottie number" ≈ 0.739). Compute $f'(x^*) = -\sin(x^*)$. Is the fixed point stable?

(d) For $f(x) = 2x(1 - x)$ on $[0, 1]$: find all fixed points and compute f' at each. Which are stable?

(e) **Why smooth structure matters here.** In Set 4 we could only say "a fixed point exists" (topology). Now we can say "the orbit converges at rate $|f'(x^*)|^n$" (smooth structure). The derivative which requires smoothness gives quantitative information that topology alone cannot.

(f) On a general smooth manifold M, the fixed point x^* of $f : M \to M$ has a well-defined derivative $df_{x^*} : T_{x^*}M \to T_{x^*}M$ (a linear map on the tangent space). The orbit is stable if all eigenvalues of df_{x^*} have absolute value less than 1. This will become the foundation of the theory of dynamical systems on manifolds.

But something is still missing: the kinetic energy $K = \frac{1}{2}g_{ij}\dot{\theta}^i\dot{\theta}^j$ introduced a matrix g_{ij} a way to measure lengths and angles on the manifold. This is a **Riemannian metric**. Without a metric, we can differentiate, but we cannot measure distances or angles. The next natural question is: what is the "shortest path" between two configurations? That is a geodesic the subject of Riemannian geometry.***Reflection.*** *Smooth manifolds gave us velocity, kinetic energy, and the rate of convergence of orbits. The equations of motion can now be written as differential equations on the configuration manifold.*

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Physics Part B: Field Theories as Invariant Machines

We now pause to look at the other direction of the physics-mathematics interaction. In Part A we used smooth manifolds to do physics. Here we ask: can physics classify *manifolds? Specifically, can a quantum field theory produce an invariant that tells us when two smooth manifolds are the same or different? The answer is yes but with a crucial limitation that turns out to be the deepest open problem in low-dimensional topology.*

The setup: what is a QFT invariant?

A **topological quantum field theory (TQFT)** is a quantum field theory whose partition function $Z(M)$ depends only on the topology of the spacetime manifold M not on any metric, coordinates, or geometric data.

The partition function is defined (heuristically) as a path integral:

$$Z(M) = \int_{\mathcal{F}} \mathcal{D}\phi \; e^{-S[\phi,M]}$$

where:

- $\mathcal{F}$ is the space of all field configurations on M
- $S[\phi, M]$ is the action a functional of the fields ϕ
- $\mathcal{D}\phi$ is the (formal) measure over all configurations

For $Z(M)$ to be a topological invariant it must satisfy two conditions:

1. $Z(M)$ does not depend on any choice of metric on M
2. $Z(M) = Z(N)$ whenever M and N are diffeomorphic

The second condition means: if two manifolds produce different partition functions, they are *definitely not diffeomorphic*. If they produce the same partition function, we cannot conclude they are diffeomorphic the theory might simply not be sensitive enough.

Observation 0.

A TQFT is an *obstruction machine*: it can certify that two manifolds are different, but it cannot certify that they are the same.

Dimension 1: supersymmetric quantum mechanics

The simplest TQFT lives on a 1-dimensional spacetime a point particle moving on a target manifold M.

The field theory. The fields are:

- A bosonic field $x \in M$ (the position of the particle)
- Two fermionic (Grassmann) fields ρ, χ satisfying $\rho\chi = -\chi\rho$

The action is the **Morse-theoretic Lagrangian**:

$$\mathcal{L} = -\frac{1}{2}s(x)^2 + \rho\,\frac{ds}{dx}\,\chi$$

where $s : M \to \mathbb{R}$ is a smooth function (the "Morse function"). The partition function is:

$$Z = \int \frac{dx}{\sqrt{2\pi}}\,d\rho\,d\chi\,\exp\left(-\frac{t^2}{2}s(x)^2 + \rho\,\frac{ds}{dx}\,\chi\right).$$

Why Z is a topological invariant. The action has a **BRST symmetry**:

$$\delta x = \chi, \qquad \delta\rho = s(x), \qquad \delta\chi = 0,$$

which satisfies $\delta^2 = 0$ (nilpotent). One can show that the action is BRST-exact: $\mathcal{L} = \delta(\cdots)$. This implies that Z is independent of the parameter t we can take $t \to \infty$ without changing Z.

Localization. As $t \to \infty$, the Gaussian $e^{-t^2 s(x)^2/2}$ forces the integral to localize at the zeros of s. Using the Grassmann integration rules $\int d\rho\, d\chi\, \rho\chi = 1$, one obtains:

$$Z = \sum_{s(x_i)=0} \operatorname{sign}\left(\left.\frac{ds}{dx}\right|_{x_i} \right) = \sum_i \pm 1.$$

The result. By the Poincaré-Hopf theorem, this sum equals the Euler characteristic:

$$\boxed{Z = \chi(M).}$$

The partition function is independent of the choice of Morse function s and depends only on M it is a topological invariant.

B1. (Computing Z in 1D.)

Let $M = S^1$ and choose the Morse function $s(\theta) = \sin\theta$.

(a) Find all zeros θ_i of s.

(b) Compute $\operatorname{sign}(s'(\theta_i))$ at each zero.

(c) Sum the signs: what is Z?

(d) Now choose a different Morse function, say $s(\theta) = \sin 2\theta$. Find the zeros, compute the signs, and sum. Do you get the same Z?

(e) Verify: $\chi(S^1) = 0$. Consistent?

B2. (Detecting non-homeomorphic 1-manifolds.)

Compute $Z = \chi(M)$ for:

(a) $M = S^1$ (compact, no boundary)

(b) $M = \mathbb{R}$ (non-compact)

(c) $M = [0, 1]$ (compact with boundary)

(d) $M = S^1 \sqcup S^1$ (disjoint union of two circles)

Which pairs are distinguished by Z? Which have the same Z?

B3. (The reverse direction fails.)

$M = \mathbb{R}$ and $N = (0, 1)$ both have $\chi = 1$, so $Z(M) = Z(N) = 1$.

(a) Are M and N homeomorphic? Prove your answer using a topological invariant from Set 4.

(b) Conclude: $Z = \chi$ cannot distinguish $\mathbb{R}$ from $(0, 1)$. The partition function gives the same value for two non-homeomorphic spaces.

(c) This shows the TQFT is not a **complete invariant** for 1-manifolds. What additional information would you need to fully classify compact 1-manifolds?

Dimension 2: Gauss-Bonnet as a field theory

The field theory. In 2 dimensions the field is a Riemannian metric g on the surface Σ. The action is the **Einstein-Hilbert action in 2D**:

$$S[g] = \frac{1}{4\pi} \int_\Sigma R \, d\mathrm{vol}_g,$$

where R is the scalar curvature of g and $d\mathrm{vol}_g = \sqrt{\det g} \, d^2x$ is the area element. The partition function is:

$$Z = S[g] = \frac{1}{4\pi} \int_\Sigma R \, d\mathrm{vol}_g.$$

Why Z does not depend on the metric. This seems paradoxical: $S[g]$ is defined using g, yet we claim it is metric-independent. The resolution is the **Gauss-Bonnet theorem**:

$$\frac{1}{4\pi} \int_\Sigma R \, d\mathrm{vol}_g = \chi(\Sigma) = 2 - 2g,$$

which holds for *any* Riemannian metric g on a closed oriented surface Σ of genus g.

Let us see why this is plausible by tracking how R and $d\mathrm{vol}_g$ change when we scale the metric. If we replace $g \to \lambda^2 g$ (a uniform rescaling by $\lambda > 0$):

$$d\mathrm{vol}_{\lambda^2 g} = \lambda^2 \, d\mathrm{vol}_g, \qquad R_{\lambda^2 g} = \frac{1}{\lambda^2} R_g.$$

Therefore:

$$\int_\Sigma R_{\lambda^2 g} \, d\mathrm{vol}_{\lambda^2 g} = \int_\Sigma \frac{1}{\lambda^2} R_g \cdot \lambda^2 \, d\mathrm{vol}_g = \int_\Sigma R_g \, d\mathrm{vol}_g.$$

The λ^2 factors cancel exactly the integral is scale-invariant. More generally, under any smooth deformation of the metric, the integral is unchanged. This is what makes Z a topological invariant.

The local version: Gaussian curvature. For a surface embedded in $\mathbb{R}^3$, the scalar curvature reduces to twice the **Gaussian curvature** K:

$$R = 2K, \qquad \text{so} \qquad Z = \frac{1}{2\pi} \int_\Sigma K \, d\mathrm{vol}.$$

The Gaussian curvature measures how curved the surface is at each point:

- $K > 0$: locally like a sphere (positively curved, like a dome)
- $K = 0$: locally flat (like a cylinder or flat plane)
- $K < 0$: locally like a saddle (negatively curved)

Gauss-Bonnet says: no matter how you embed or distort the surface, the *total* curvature $\int K \, d\mathrm{vol}$ always equals $2\pi\chi(\Sigma)$. Positive and negative curvature must balance to give the right total.

Deriving the partition function on S^2. The round sphere of radius r has constant Gaussian curvature $K = 1/r^2$ and area $4\pi r^2$. Therefore:

$$Z(S^2) = \frac{1}{2\pi} \cdot \frac{1}{r^2} \cdot 4\pi r^2 = 2 = \chi(S^2). \checkmark$$

Notice the r^2 cancels the result is independent of the radius.

Deriving the partition function on T^2. The flat torus has $K = 0$ everywhere (it is locally isometric to $\mathbb{R}^2$). Therefore:

$$Z(T^2) = \frac{1}{2\pi} \int_{T^2} 0 \, d\text{vol} = 0 = \chi(T^2). \checkmark$$

Deriving the partition function on Σ_g. For a surface of genus g, one can put a metric of constant curvature $K = \text{sign}(2 - 2g)$:

- $g = 0$ (S^2): constant $K = +1$ metric (round sphere)
- $g = 1$ (T^2): constant $K = 0$ metric (flat torus)
- $g \geq 2$: constant $K = -1$ metric (hyperbolic surface)

In each case, $\int_{\Sigma_g} K \, d\text{vol} = K \cdot \text{Area}(\Sigma_g)$, and the area is determined by K and χ via Gauss-Bonnet. For $K = -1$: $\text{Area}(\Sigma_g) = 2\pi|\chi(\Sigma_g)| = 2\pi(2g - 2)$.

B4. (Explicit derivation on S^2 with a non-round metric.)

Consider S^2 with a different metric not round, but bumpy. In a neighborhood of the north pole, suppose the curvature is $K = 2$ (more curved than round), and near the equator $K = 1/2$ (less curved).

(a) Without computing anything, what must $\frac{1}{2\pi} \int_{S^2} K \, d\text{vol}$ equal? Why?

(b) If the bumpy metric gives curvature $K = 2$ on a region of area A_1 and $K = 1/2$ on the rest (area $A_2 = 4\pi - A_1$ approximately), what constraint does Gauss-Bonnet place on A_1?

(c) This shows that if you make one part of the sphere more curved, another part must become less curved to compensate. What is the physical analogy? *(Think of a balloon: squeezing one end makes the other bulge.)*

B5. (Curvature must average to the right value.)

(a) Show that the **average curvature** of a closed oriented surface Σ_g satisfies:

$$\langle K \rangle = \frac{\int_{\Sigma_g} K \, d\text{vol}}{\text{Area}(\Sigma_g)} = \frac{2\pi\chi(\Sigma_g)}{\text{Area}(\Sigma_g)}.$$

(b) What does this say for T^2? Is it possible to have $K > 0$ everywhere on T^2?

(c) What does this say for Σ_2 (genus 2)? Can $K > 0$ anywhere? Must $K < 0$ somewhere?

(d) **Physical interpretation.** A physical surface (like a soap film) satisfying the Euler-Lagrange equations for the action $S[g] = \int K \, d\text{vol}$ would have to distribute its curvature to match the topological constraint $\int K = 2\pi\chi$. The topology of the surface constrains the physics of the field living on it.

B6. (The partition function as a topological invariant explicit check.)

Take the torus T^2 and give it two different metrics:

- g_1: the flat metric (embedding as a product $S^1 \times S^1$ with equal radii $r = 1$), with $K_1 = 0$ everywhere.
- g_2: the induced metric from the standard embedding of T^2 in $\mathbb{R}^3$ as a surface of revolution with major radius $R = 2$, minor radius $r = 1$. In this metric:

$$K_2(\theta) = \frac{\cos\theta}{r(R + r\cos\theta)} = \frac{\cos\theta}{2 + \cos\theta}$$

where θ is the angle around the tube.

(a) For g_1: compute $\frac{1}{2\pi}\int_{T^2} K_1 \, d\text{vol}_{g_1}$.

(b) For g_2: the area element on the embedded torus is $d\text{vol}_{g_2} = r(R + r\cos\theta)\, d\theta\, d\phi$. Compute:

$$\frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} \frac{\cos\theta}{2 + \cos\theta} \cdot (2 + \cos\theta)\, d\theta\, d\phi.$$

Hint: simplify the integrand first. What is $\int_0^{2\pi} \cos\theta \, d\theta$?

(c) Both metrics give $Z = 0 = \chi(T^2)$. Verify this and explain why it had to be so regardless of which metric you chose.

(d) **Key lesson.** The partition function $Z = \chi(\Sigma_g)$ does not depend on *how* you compute it (which metric you use) only on the topology of Σ_g. This is what makes it a topological invariant: the field (metric) is arbitrary, the result is not.

B7. (The invariant is complete in 2D and the reverse direction holds.)

(a) Compute $Z(\Sigma_g) = \chi(\Sigma_g)$ for $g = 0, 1, 2, 3$. Are any two the same?

(b) For closed oriented surfaces, the classification theorem says:

$$\Sigma_g \cong \Sigma_h \iff g = h \iff \chi(\Sigma_g) = \chi(\Sigma_h).$$

This means $Z = \chi$ is a **complete invariant** in 2D: the forward direction ($\Rightarrow$) and the reverse ($\Leftarrow$) both hold.

(c) Compare with 1D: why did the reverse direction fail there ($\mathbb{R}$ and $(0, 1)$ both had $Z = 1$)? What is different about closed oriented surfaces that makes χ complete?

(d) **Important caveat.** The completeness in 2D is a theorem about closed *oriented* surfaces. For non-orientable surfaces (like $\mathbb{R}P^2$ or the Klein bottle), χ alone is not enough you also need the orientability. Compute $\chi(\mathbb{R}P^2)$ and $\chi(T^2)$. Are they the same? Are the surfaces homeomorphic?

B8. (Summary: the invariant machine in dimensions 1 and 2.)

Dim	Field theory	Invariant	Detects difference?	Complete?
1	SQFT / Morse	$\chi(M)$	S^1 vs $S^1 \sqcup S^1$: yes $\mathbb{R}$ vs $(0,1)$: no	No
2	Gauss-Bonnet	$\chi(\Sigma_g)$	S^2 vs T^2: yes (for closed oriented surfaces)	Yes

(a) In 1D, the TQFT is not complete. What extra information beyond χ would classify all 1-manifolds (compact and non-compact, with and without boundary)?

(b) In 2D, Gauss-Bonnet is complete for closed oriented surfaces. The proof of completeness is the **classification of surfaces** a purely topological theorem, not a physics result. What role does the field theory play: does it prove completeness, or does it just compute the invariant?

(c) **The key distinction, stated cleanly:**

Field theory	Classical topology
Produces the invariant	Proves it is complete
Detects differences	Proves no differences were missed
Computes χ	Classifies manifolds using χ

Neither can do the other's job.

In both cases, the physics gives us the invariant. Whether that invariant is complete whether equal values guarantee the spaces are the same is a question for mathematics, not physics. In 2D, we are lucky: χ happens to be complete for closed oriented surfaces. In higher dimensions, this luck runs out, and the search for complete invariants remains one of the deepest open problems in mathematics.***Final reflection on Part B.*** *A quantum field theory produces a number $Z(M)$ the partition function. In 1D (supersymmetric quantum mechanics), the partition function localizes at critical points of a Morse function and computes $\chi(M)$. In 2D (Gauss-Bonnet), the partition function is the total curvature, which is metric-independent and equals $\chi(\Sigma_g) = 2 - 2g$.*

In both cases, the physics gives us the invariant. Whether that invariant is complete whether equal values guarantee the spaces are the same is a question for mathematics, not physics. In 2D, we are lucky: χ happens to be complete for closed oriented surfaces. In higher dimensions, this luck runs out, and the search for complete invariants remains one of the deepest open problems in mathematics.